MALEK BENCHEHIDA

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EDUCATION

University of Illinois Chicago Chicago, IL

Bachelor of Science in Computer Science **Expected December 2026**

- **Leadership:** Technical Software Lead, SAE Formula Electric Team; engineered telemetry pipelines monitoring real-time vehicle speed and throttle-pedal position (0–100%) to optimize race performance.

SOFTWARE ENGINEERING EXPERIENCE

MatchNice Chicago, IL (Remote)

Software Engineer **June 2026 – Present**

- Engineered a scalable donation-matching platform replacing a legacy no-code prototype, driving production launch alongside CTO by architecting a robust full-stack Next.js, TypeScript, and Supabase PostgreSQL solution.
- Enabled milestone-based fund matching for a recurring-donation system by building responsive React campaign controls, Express REST APIs, and PostgreSQL persistence layers, contributing directly to vNext integration.
- Guaranteed 100% data integrity during Fundraise Up and GoFundMe webhook ingestion by designing background sync workers that validate provider event IDs and enforce database uniqueness constraints during retries.
- Accelerated nonprofit research and sales-playbook turnaround by 98.6% (18 hours to 15 minutes) by automating URL intake and document synthesis using Google Apps Script, Google Forms, and a Google Gem.

CAS Enterprise Inc. Chicago, IL

DevOps & Full Stack Developer **Aug 2023 – June 2026**

- Decreased client-side payload size by 30% by optimizing complex GraphQL queries, eliminating redundant fields, and migrating initial data fetching to Next.js server-side rendering architecture.
- Accelerated frontend performance to a 99/100 Lighthouse score and cut Largest Contentful Paint from 3.4s to 1.2s by combining server rendering, next/image preloading, and dynamic JavaScript imports.
- Executed application infrastructure migration from AWS to Azure, provisioning and configuring highly available App Service, Blob Storage, and Active Directory environments for enterprise reliability.
- Reduced deployment latency by 91% (from 2 hours to 10 minutes) by engineering automated CI/CD build and release workflows using Python, significantly accelerating engineering release cycles.
- Eliminated recurring backend out-of-memory crashes by tracing oversized SQL queries returning 5,235 unpaginated rows, injecting targeted database indexes, and batching high-throughput data streams.

PROJECTS

EdgeGuardNano — Edge Telemetry & Inference **2026**

- Enabled deterministic, offline telemetry processing on an NVIDIA Jetson AGX Thor by engineering asynchronous, non-blocking ingestion pipelines in Rust, leveraging strict ownership for memory safety.
- Optimized LightGBM inference pipelines for ARM64 CPUs via advanced profiling; surfaced real-time edge-node metrics and DDoS classifications on a Next.js dashboard using GraphQL subscriptions.

P2P Crypto Wallet Prototype **2025**

- Protected transaction consistency during concurrent transfers in Go and Node.js by enforcing strict PostgreSQL ACID compliance and SELECT FOR UPDATE row locks; validated fault tolerance with automated rollback tests.
- Secured wallet access and state mutations by implementing BIP39 mnemonic key generation and a strict TOTP-based multifactor authentication pipeline to safeguard user cryptographic keys.

Real-Time Gesture Inference Engine **2025**

- Built a low-latency webcam-to-prediction pipeline integrating MediaPipe, a Python FastAPI endpoint, and a scikit-learn random forest model to drive real-time gesture classifications for a JavaScript application.

RESEARCH EXPERIENCE

Chicago Tech Circle Chicago, IL

Machine Learning Research Collaborator **Sept 2026 – Present**

- Reviewing chest X-ray and visceral-fat literature to guide feature engineering for an early-stage Python/PyTorch study predicting type 2 diabetes risk based on multi-modal clinical biomarkers.

TECHNICAL SKILLS

- **Languages:** Rust, Go, Python, TypeScript, JavaScript, SQL, Java, C++, C, HTML5, CSS3, Bash
- **Backend & APIs:** Node.js, Express.js, FastAPI, GraphQL, REST APIs, OpenAPI, Drizzle ORM, Zod, WebSockets
- **Databases & Infrastructure:** PostgreSQL, AWS, Azure, Docker, Linux, Supabase, MongoDB, Redis, CI/CD
- **Delivery & Tooling:** GitHub Actions, Git, Linux/Unix Shell, Postman, Jest, Vite, Google Apps Script
- **Frontend & ML:** React, Next.js, Tailwind CSS, scikit-learn, LightGBM, MediaPipe, PyTorch, NumPy, Pandas